SITUATIONAL AI

Finance & Economics Intelligence Framework

SCBCM Models for Accounting, Risk, Fraud, Compliance, and Macroeconomic Stability

Sai Jagadish Kumar Manchikanti · Dr. Ramesh Subramoniam · Prasad Pillai, CEO Mindspace AI

PRESENTATION OVERVIEW

01 **Research Questions**

What problems are we solving?

02 **What is Situational AI?**

SCBCM framework explained

03 **Applications in Finance**

Fraud, risk, compliance, systemic stability

04 **Research Gaps**

What's missing in the literature?

05 **Path Forward**

How to advance this research

06 **Future Impact**

Why this matters

RESEARCH QUESTIONS

What Are We Trying to Solve?

RQ1	How can SCBCM-based AI models enhance the detection and interpretation of situational risks in accounting, fraud, compliance, and cash-flow management at the firm level?
RQ2	Which behavioral and contextual patterns are most predictive of distress, misreporting, or non-compliance — and how do these aggregate at sectoral or macroeconomic levels?
RQ3	In what ways do AI-generated financial risk signals alter information asymmetry, capital allocation, and the cost of capital in financial markets?
RQ4	How can situational AI insights be incorporated into macro-prudential frameworks to reduce the likelihood or severity of financial crises?

WHAT IS SITUATIONAL AI?

A financial event is not just data

it is an occurrence situated within behavioral, temporal, network, and environmental context simultaneously.

Situation = Signals + Context + Memory → Actionable Meaning

Unlike traditional AI that classifies raw features, Situational AI asks: anomalous given WHO, WHEN, WHERE, and UNDER WHAT CONDITIONS?

SCBCM's 4 Contextual Layers

Behavioral

Historical activity profiles & deviations

Temporal

Intraday, quarterly & macro-financial cycles

Network

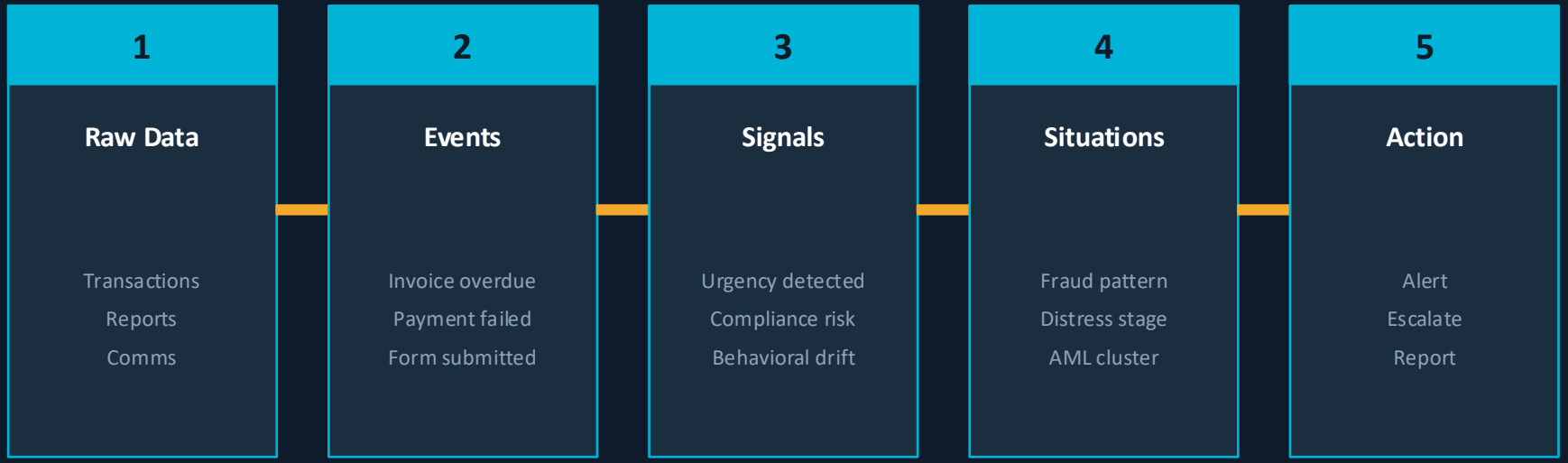
Dynamic relationship graphs (GNN-based)

Environmental

Macro conditions, sector, regulatory climate

THE SCBCM ARCHITECTURE

Situation-Centered Behavioral & Context Modeling

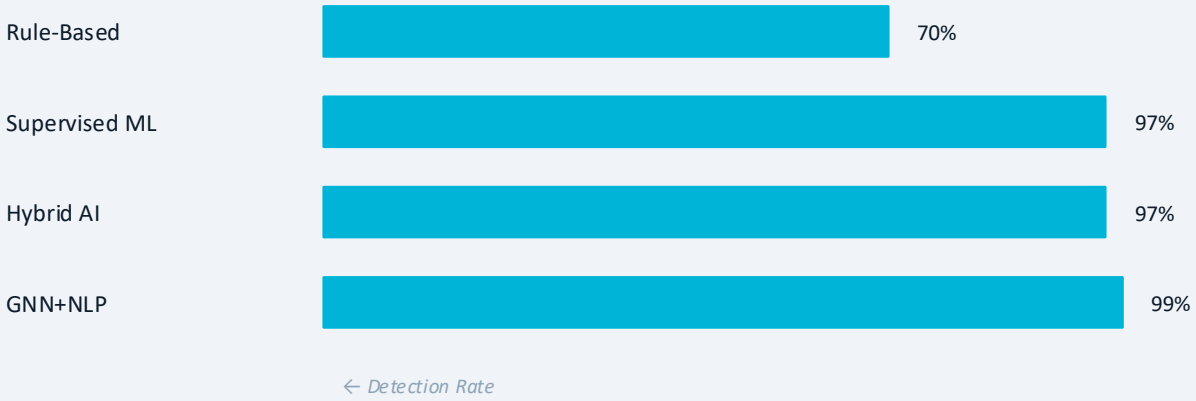


APPLICATION 1

Fraud Detection & AML Compliance

95–99%	<4%	70–85%	$200B+
Detection rate (Hybrid AI)	False positive rate vs. 95% legacy	Cost reduction via continuous audit	Global AML cost that Situational AI cuts

Model Performance Comparison

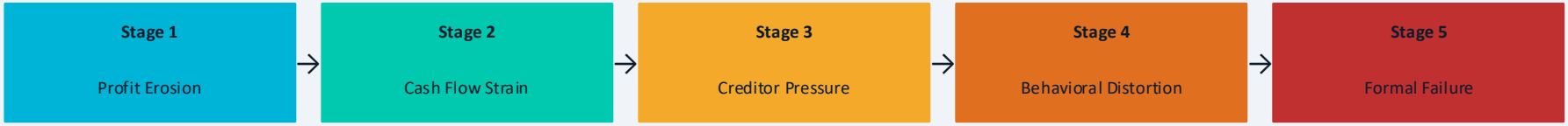


Key Insight

Graph Neural Networks detect 30–50% more suspicious patterns by modeling multi-party relationships — catching triangular payment schemes invisible to rule-based engines.

APPLICATION 2

Corporate Distress Prediction & Firm Behavior



⟵ Situational AI Early Warning Window: 6–24 months ⟶

Prediction Accuracy by Model

Model	1-Year	2-Year	3-Year
Altman Z-Score	72–80%	68–72%	58–64%
Random Forest	88–92%	84–88%	78–82%
SCBCM Hybrid	**91–95%**	**88–93%**	**83–88%**

Why SCBCM Wins at Longer Horizons

Classical models degrade to near-random performance at 3–5 years. SCBCM retains AUC > 0.85 by combining behavioral signals with macro-financial context — giving creditors and boards actionable time to intervene.

APPLICATIONS 3 & 4

Information Asymmetry & Macroprudential Stability

Information Asymmetry & Market Efficiency

- AI-derived text measures (BERT/transformers) analyze disclosures at scale
- Predict cost of capital with 15–25% greater explanatory power than traditional proxies
- Context matters: the same language carries different weight depending on industry, timing, analyst coverage
- Regulators can calibrate disclosure rules using heterogeneous effects for the first time at scale

Macroprudential & Systemic Risk

- Network-based DebtRank + GNN models provide continuous (vs quarterly) systemic risk scoring
- AUC > 0.85 at 2-year horizon for banking crisis early warning
- Central banks (IMF/BIS) operationally use NLP to monitor financial news in real time
- 30–50% more AML detection of suspicious activity patterns vs legacy rule engines

RESEARCH GAPS

Emerging Market Evidence

Critical

Nearly all studies use US, EU, or Japanese data. Emerging markets have higher data sparsity, weaker audit quality, and more regime changes — making direct application unreliable.

Adversarial Robustness

Critical

Fraud models are deployed against intelligent adversaries who adapt. How quickly can fraudsters identify detection signatures? What retraining cadence preserves performance?

Macroeconomic Transmission Effects

High

Does widespread AI adoption create correlated behavior and herding that amplifies systemic risk? The system-level economic effects of SCBCM adoption remain largely unquantified.

Regulatory Design Interactions

High

Explainability requirements cost 2–5% AUC accuracy. Capital calibration based on AI scores creates feedback loops whose stability properties are analytically unresolved.

LITERATURE COVERAGE MAP

Where the Research Is — and Where It Isn't



Red bars = critical gaps requiring priority research investment

HOW TO PROGRESS THIS STUDY

Phase 1

Design & Prototype SCBCM Models

- Build SCBCM detection models for fraud, compliance, cash-flow warnings
- Validate on real firm-level financial datasets (US + emerging markets)
- Benchmark against classical models (Altman Z, Ohlson, Random Forest)

Phase 2

Aggregate to Macro Level

- Aggregate firm-level situational signals into sectoral risk indicators
- Model transmission: how firm-level distortions scale to systemic fragility
- Test information asymmetry effects on cost of capital using AI-derived measures

Phase 3

Policy & Regulatory Integration

- Propose implementation pathways for embedding SCBCM in supervisory tools
- Design macro-prudential early-warning systems with 12–24 month lead times
- Address adversarial robustness via adversarial training & retraining cadences

METHODOLOGY

Research Design at a Glance

Data Sources

Financial statements, transaction logs, earnings call transcripts, regulatory filings, news feeds

Firm-Level Models

SCBCM hybrid ensembles: GNN + NLP + gradient boosting, validated across sectors

Macro-Level Models

Network contagion models (DebtRank), credit cycle early warning, CoVaR / SRISK

Benchmarks

Altman Z-Score, Ohlson O-Score, single-method ML baselines, rule-based systems

Evaluation Metrics

AUC, detection rate, false positive rate, cost of capital R^2, crisis prediction lead time

Explainability

SHAP attribution + counterfactual explanations to satisfy SOX, PCAOB, ECB standards

THE FUTURE IMPACT

For Firms

- Real-time fraud & distress warnings replace periodic audits
- Behavioral AI strengthens internal controls beyond rule-based compliance
- Boards get months more lead time to act before distress becomes irreversible

For Markets

- Higher disclosure quality → lower cost of capital for transparent firms
- Reduced information asymmetry improves capital allocation efficiency
- AI-driven analyst tools level the playing field for under-covered firms

For Regulators

- Continuous macroprudential monitoring replaces lagging quarterly reports
- Countercyclical buffers calibrated with 12–18 months additional lead time
- System-level contagion modeled in real time, enabling faster policy response

THE BIGGER PICTURE

From Detecting Events to Understanding Situations

The fundamental shift Situational AI enables:

Static, retrospective audits	→	Continuous, forward-looking surveillance
✕ Rule-based, brittle controls	→	✓ Behavioral, adaptive intelligence
✕ Institution-level risk silos	→	Network-aware systemic risk modeling
Manual, sample-based review	→	✓ 100% transaction coverage in real time

KEY TAKEAWAYS

Situational AI is not a small step — it's a new Standard.

1. SCBCM outperforms every conventional model across fraud, distress, compliance, and systemic risk
2. Context is the missing ingredient in financial AI — behavioral + temporal + network + environmental
3. Critical gaps remain: emerging markets, adversarial robustness, macro transmission effects
4. This research bridges firm-level governance and macro-financial stability through a single unified lens